3. **What are the major differences between the same devices from different brands?**

   The major differences typically lie in performance, features, software, scalability, and cost. For example:

   - Routers: Cisco ISR 4321 provides strong enterprise-grade security and stable throughput, Juniper SRX345 offers higher scalability and redundancy, and TP-Link TL-ER7206 is more cost-effective but limited in features.
   - Switches: Cisco Catalyst 9300 and 9500 series differ in throughput, Layer 3 capabilities, and port configurations. Higher-end switches offer greater reliability, advanced features, and faster uplinks, while lower-end or competitor models may be sufficient for smaller deployments but lack advanced routing options.
   - Wireless Access Points: Cisco 1850/9115AX supports WiFi 6 and higher user capacity, Ubiquiti UAP-AC-Pro offers moderate speed and coverage, while TP-Link EAP245 is cheaper but supports fewer concurrent users.
   - Servers: Dell PowerEdge R550 provides high RAM capacity, dual CPU support, and virtualization readiness; HPE ProLiant and Lenovo ThinkSystem offer lower-cost options with limited expansion.
   - Printers: HP, Canon, and Brother devices differ in speed, energy efficiency, and reliability, affecting suitability for high-usage environments like student lounges.

   Overall, the differences reflect trade-offs between cost, performance, scalability, and advanced features, making brand and model selection crucial depending on the lab or network requirements.

Reflection Task 3

By comparing networking devices from different brands, we learned that price differences usually reflect performance, reliability, security features, and scalability. Enterprise-level networking devices are more expensive, but they provide better stability and long-term support, which are important for a faculty network. However, we also realised that the most expensive option is not always the best choice for an educational environment.Through this task, we learned to consider cost-effectiveness when selecting devices. With a limited budget, it is necessary to balance performance, reliability, and price to meet actual requirements. This process helped us better understand how device selection affects the overall network design and budget planning.

TASK 4

4.1 Identify the Work Area on the Floor Plan

4.1.1 Ground floor

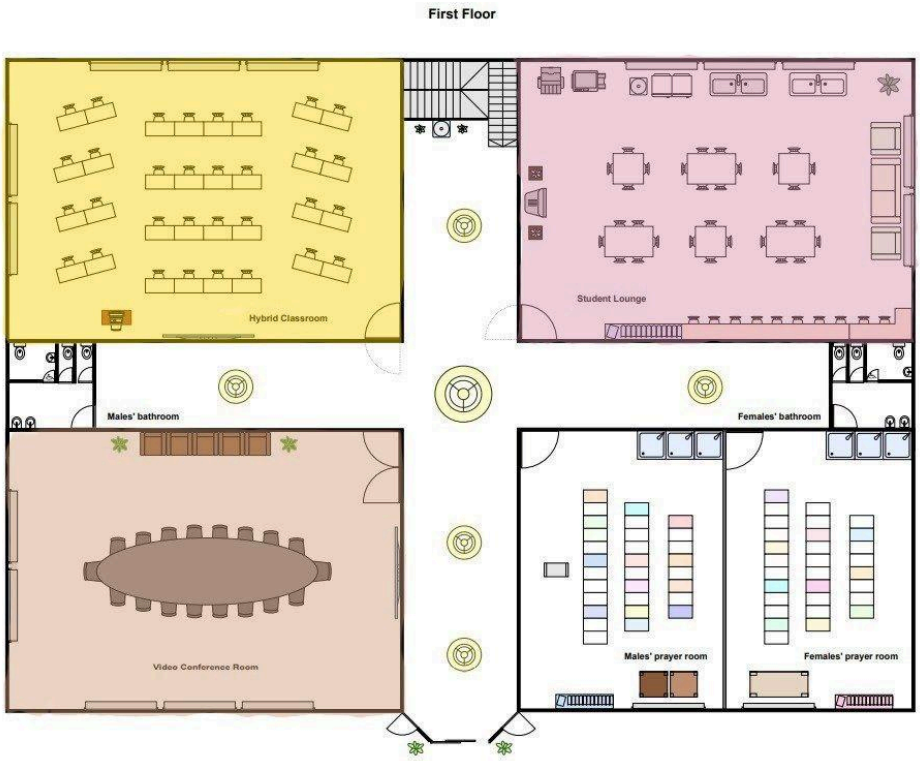


Figure 4.1.1

The ground floor work areas are designed to support teaching, collaboration, communication and student activities while ensuring reliable network connectivity. The hybrid classroom is equipped with desktop or lecturer PCs, interactive smart boards, projectors, wireless microphones and video conferencing cameras. These devices are connected using structured Cat 6 Ethernet cabling to access switches located in the network cabinet, ensuring stable wired connectivity for teaching equipment while wireless access points (APs) provide internet access for student's laptops and mobile devices. The cabling is routed through ceiling trunking to maintain a clean layout and allow future expansion.

The video conference room functions as a collaboration and communication workspace that requires high network reliability. It is equipped with high-definition cameras, conference

microphones, display screen and a dedicated conferencing PC. All critical devices are connected via wired ethernet connections to minimize latency and packet loss during live meetings. A dedicated network switch port and quality-of-service(QoS) configuration ensure smooth audio and video transmission while proper cable management improves safety and maintenance.

A student lounge serves as an informal work and study area where students use personal laptops and mobile devices. This area mainly relies on wireless networking, supported by strategically placed access points to provide full coverage and sufficient bandwidth. Minimal wired cabling is required except for network uplinks to the access points and charging stations. This design allows flexibility, mobility and ease of use while maintaining secure network access.

4.1.2 First floor

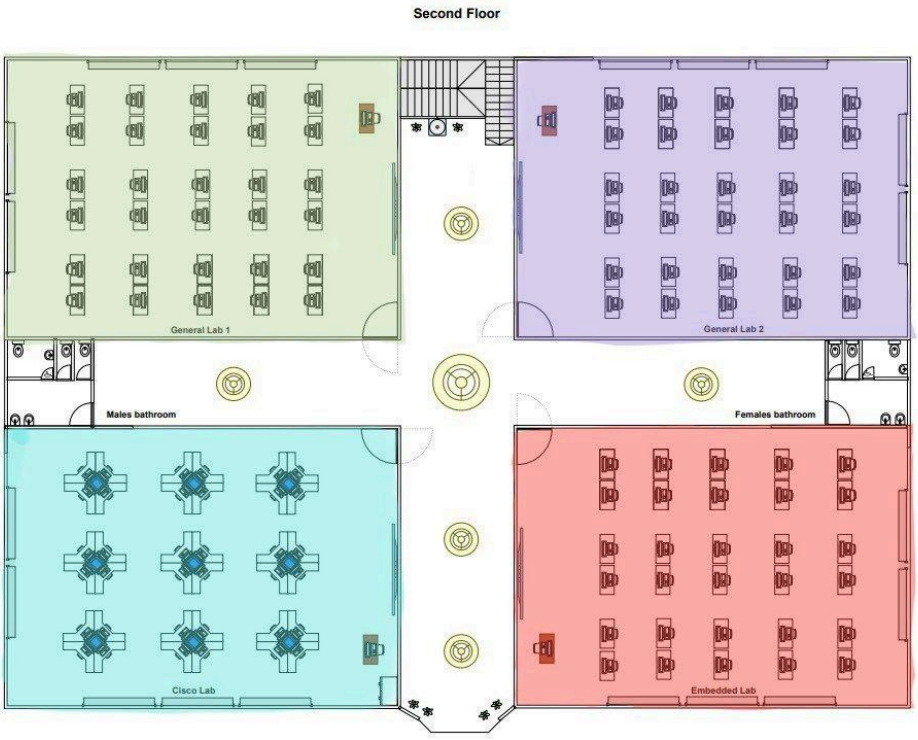


Figure 4.1.2

The first floor work areas are dedicated to technical and laboratory-based learning, requiring structured cabling and high-performance network devices. The Cisco lab is a specialized networking workspace equipped with multiple desktop computers, Cisco routers, switches, patch panels and racks. Each workstation is connected using cat 6 ethernet cables to access switches, allowing students to perform routing, switching and network configuration exercises. Patch panels are used to organize cabling and simplify troubleshooting while lab follows a star topology for efficient network management.

The embedded lab supports hardware-based and IoT-related development activities. It is equipped with computers, microcontrollers, development boards, sensors and testing instruments. Workstations are connected through wired ethernet connections for stable data transfer while wireless connectivity is provided for IoT device testing. Proper power outlets, network points and cable trunking are arranged to ensure safety and ease of maintenance during hardware experimentation.

The general labs are designed for programming, software development and general computing tasks. Each lab contains approximately 30 desktop computers connected to a local switch using structured ethernet cabling. The switch uplinks to the main network to provide high-speed internet access. Wireless access points are also installed to support additional devices such as laptops. This combination of wired and wireless connectivity ensures reliability, flexibility and scalability for various academic activities.

4.2 The Distribution and Connection of Devices

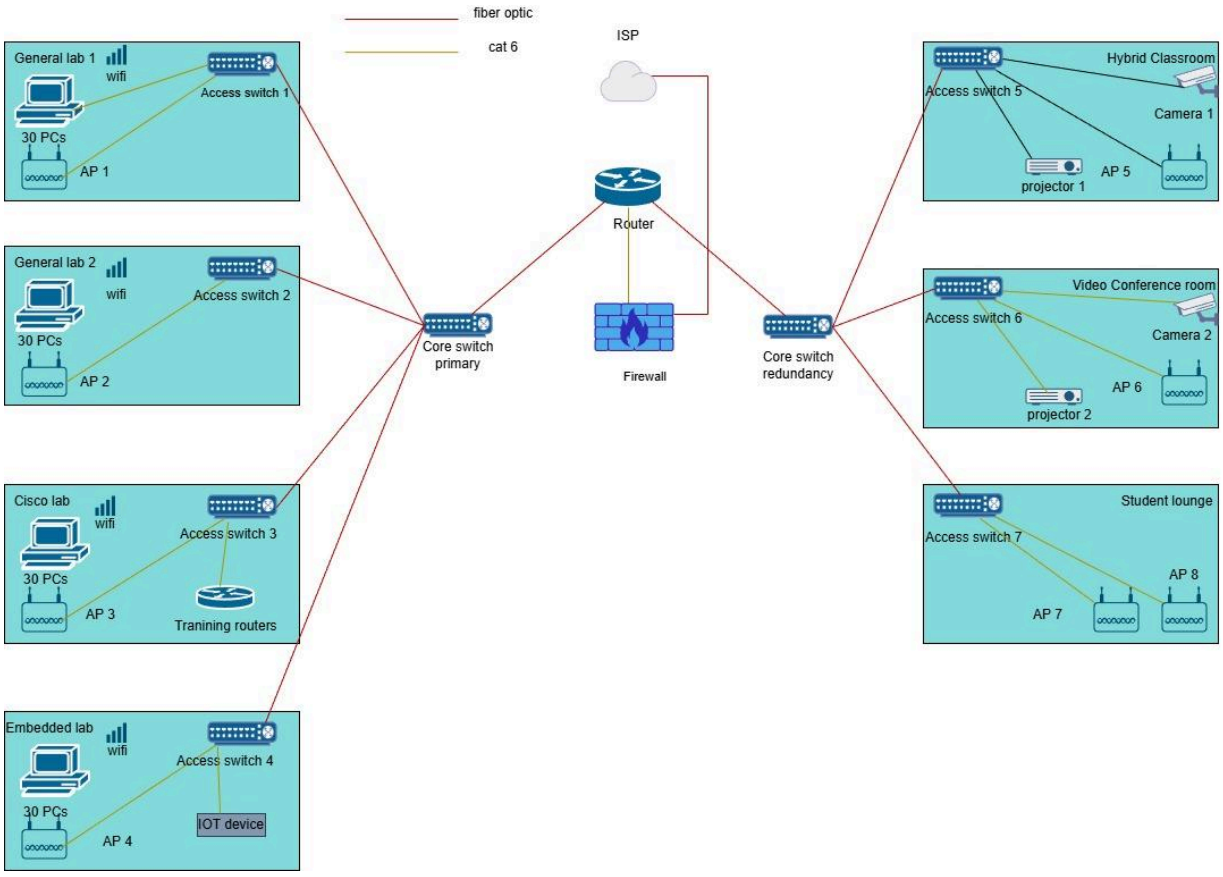


Figure 4.2.1

The network is designed to connect all rooms on both floors to the internet in a clear and organized way. Each room uses an access switch to connect end devices such as PCs, wireless access points, cameras, projectors, training routers, and IoT devices using Cat 6 cables.

All access switches are connected to the core switches using fiber optic cables to support higher bandwidth communication between floors. Two core switches are used to improve network reliability. A firewall is included in the network design to provide security control for internal network traffic. It is used to protect internal resources, and the ISP provides internet and external network services for the building.

4.2.1 Ground floor

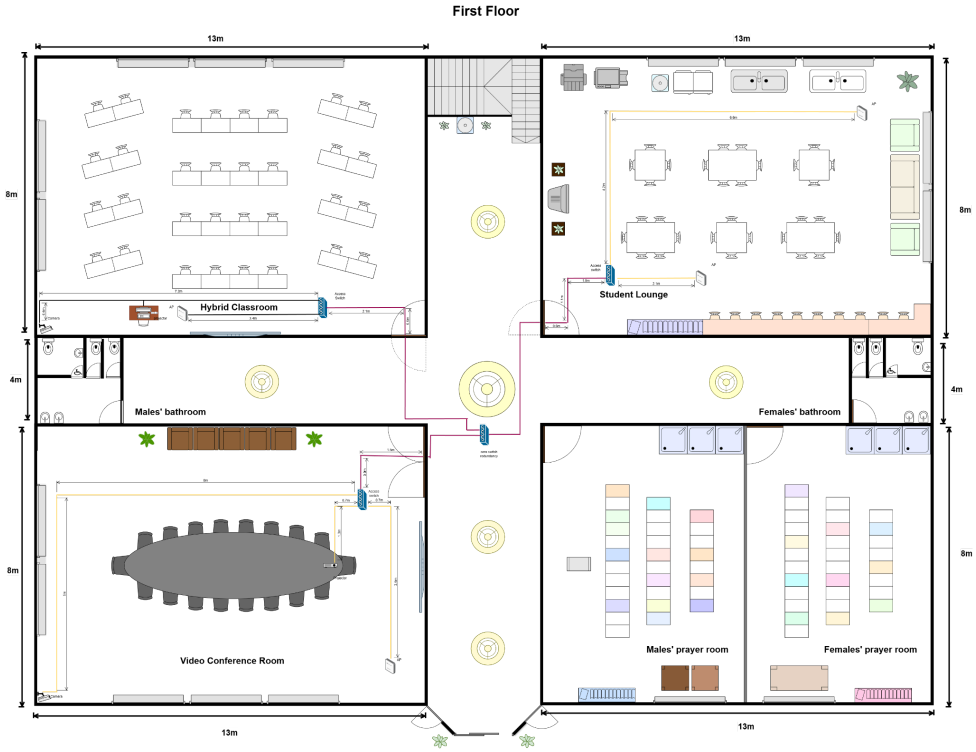


Figure 4.2.2

Backbone Cabling

The backbone cabling in this network design is used to interconnect the main network distribution points and to support high-speed data transmission across the building. Fibre optic cables are selected as the backbone medium due to their high bandwidth capacity, low signal attenuation, and suitability for handling heavy network traffic.

Horizontal Backbone Cabling

Horizontal backbone cabling refers to the fiber optic connections that run across the same floor to link the Main Distribution Facility (MDF) to major network distribution paths. In the floor plan, horizontal backbone cabling is represented by red lines running along the corridors and central areas. These fiber optic cables distribute network connectivity from the MDF to different sections of the floor, ensuring efficient data flow to all work areas.

The horizontal backbone supports high network traffic generated by laboratories, classrooms, and collaboration spaces. It also reduces congestion by acting as the main high-capacity pathway before connections are distributed through CAT6 horizontal cabling to end devices.

Vertical Backbone Cabling

Vertical backbone cabling refers to the fiber optic connections that link network distribution facilities between different floors. In this design, vertical backbone cabling connects the MDF on the ground floor to the distribution points on the upper floor. This connection allows network services to be shared across floors while maintaining high performance and reliability.

Fiber optic cabling is used for vertical backbone connections because it supports long-distance transmission and provides scalability for future network expansion.

4.2.2 First Floor

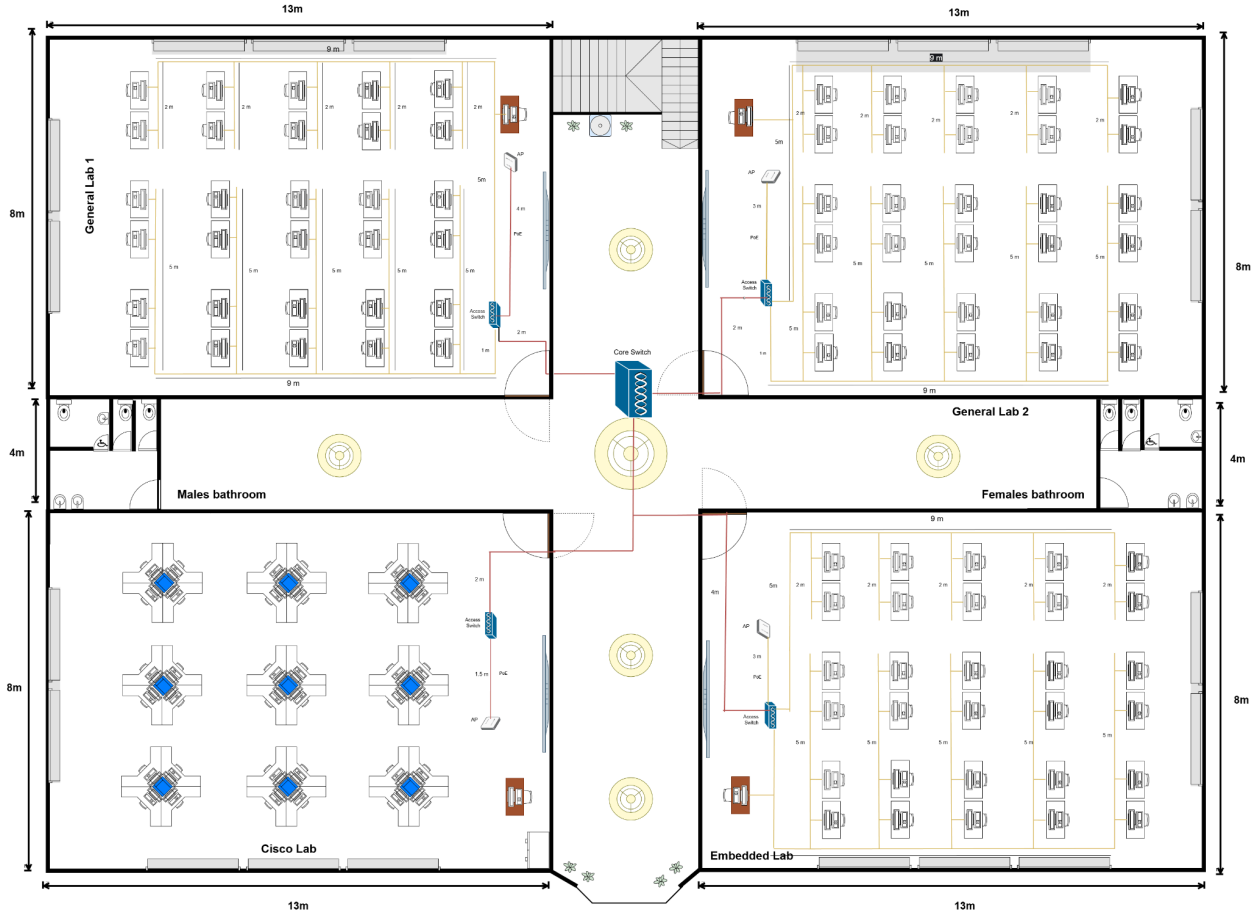


Figure 4.2.3

Main Cabling Network

The first-floor main cabling network is designed to connect all labs on the floor to the main network infrastructure. Fiber optic cables are used as the main network to ensure high-speed data transmission, low signal loss, and reliable performance for intensive lab use.

Horizontal Main Cabling Network

The first-floor horizontal main cabling network refers to the fibre optic connections that link the access switches in each lab (General Lab 1, General Lab 2, Cisco Lab, and Embedded Systems Lab) to the central switch.

These fibre optic connections run along the corridors and act as high-capacity paths, efficiently distributing network connectivity throughout the floor.

The horizontal main cabling network ensures a smooth flow of data from all labs before it is aggregated at the central switch and routed to the central network devices.

Horizontal Access Cables

Dual-stranded Cat6 cables are used to connect the horizontal access cables within each lab on the first floor. These cables connect desktop computers and wireless access points directly to the local access switch in each lab, providing a stable, high-speed network connection for peripheral devices

Vertical Backbone Cables

Vertical backbone cables connect the network infrastructure on the first floor to the existing ground floor network. Optical fiber is used for this connection to support long-distance transmission, high bandwidth, and future scalability between floors.

4.3 Number of Connections, Patch Cords, and Switch Ports Needed

Each wired device such as computers, smart boards, video conferencing equipment, routers, switches, and wireless access points is considered one network connection. For every network connection, two patch cords are required which is one connecting the device to the wall outlet and another connecting the patch panel to the switch. In addition, each connection occupies one switch port. The calculation is performed separately for each room and laboratory on the ground floor and first floor to ensure accurate network planning, efficient switch selection, and sufficient capacity for future expansion. The results of this analysis are then used to label the floor plan and support the overall network cabling design.

4.3.1 Ground Floor

Number of connection:

Area	Number of connection needed
Hybrid classroom	3
Student lounge	3
Video conference room	3

Number of patch cords:

Area	Number of patch cords needed
Hybrid classroom	6
Student lounge	6
Video conference room	6

Number of switch ports:

Area	Number of switch ports needed
Hybrid classroom	3
Student lounge	2
Video conference room	3

Ground floor total:

Items	Number of quantity needed
Connections	9
Patch cords	16
Switch ports	8

4.3.2 First Floor

Number of connection:

Area	Number of connection needed
General Lab 1	31
General Lab 2	31
Cisco Lab	45
Embedded Lab	30

Number of patch cords:

Area	Number of patch cords needed
General Lab 1	62
General Lab 2	62
Cisco Lab	90
Embedded Lab	60

Number of switch ports:

Area	Number of switch ports needed
General Lab 1	31
General Lab 2	31
Cisco Lab	45
Embedded Lab	30

First floor total:

Items	Number of quantities needed
Connections	137
Patch cords	274
Switch ports	137

Overall building total:

Items	Number of quantities needed
Connections	146
Patch cords	290
Switch ports	145

4.4 Identify Cable Types and Lengths

4.4.1 Ground Floor

a) Total Length of Cat6 Ethernet Cables needed on the Ground Floor:
   Hybrid Classroom : 7.2 + 0.6 + 0.2 + 3.4 = 11.4m
   Student Lounge : 6.6 + 4.2 + 2.1 = 12.9m
   Video Conference Room : 8 + 5 + 0.3 + 1.3 + 0.7 + 0.7 + 2.5 = 18.5m
   Total = 42.8m

b) Total Length of Fibre Optic Cables needed on the Ground Floor:
   Hybrid Classroom : 2.1 + 0.6 = 2.7m
   Student Lounge : 1 + 1.1 + 0.6 = 2.7m
   Video Conference Room : 0.9 + 1.5 = 2.4m
   To connect with the Switch core that is on the Ground floor = 10m
   Total = 17.8m

4.4.2 First Floor

a) Cat6 Ethernet Cables
   general lab1 = (9*2) + (5*5) + (2*5) + (5) + (4) + 1 = 63m .
   general lab2 = (9*2) + (5*5) + (2*5) + (5) + (3) + 1 = 62m.
   embedded lab= (9*2) + (5*5) + (2*5) + (5) + (3) + 1 =62m.
   Cisco lab = 1.5 m
   Total = 188.5 m

b) Fibre Optic Cables
   general lab1 = 2m.
   general lab2 = 2m.
   embedded lab = 4m.
   Cisco lab = 2m.
   To connect with the Switch core that is on the First floor = 10m
   Total = 20m.

Reflection Task 4

Through this task, we gained an understanding of how network design is planned based on a building's floor layout. This task required us to identify work areas and decide suitable device placement, cabling paths and network connections for both floors. By analysing different spaces, we learned how network requirements depend on usage and how this affects cabling types, switch placement and connection distribution. This task helped us realise that a well planned physical layout is important to ensure network efficiency, safety and future scalability.

This task also allowed us to reflect on the practical aspects of network implementation such as estimating the number of connections, patch cords, switch ports and cable lengths needed. We are more aware of how small design decisions can impact cost, maintenance and performance. Overall, Task 4 strengthened our understanding of physical network design and improved our ability to plan a realistic and organized network infrastructure.

TASK 5

5.0 Details of subnetting and IP assignation to each lab and room

5.1 Subnet Mask

The network address assigned by the lecturer is:

Network Address: 172.19.0.0/20

Subnet Mask: 255.255.240.0

Host Capacity Calculation

Subnet	Area	IP Address/ Subnet Mask (Binary)	IP Address/ Subnet Mask (Decimal)	Usable Account (Host)
0	Student Lounge (WiFi)	10101100.00010011.00000000.00000000 / 11111111.11111111.11111111.00000000	172.19.0.0/24	256
1	Hybrid Classroom	10101100.00010011.00000001.00000000 / 11111111.11111111.11111111.10000000	172.19.1.0/25	126
2	General Lab 1	10101100.00010011.00000001.10000000 / 11111111.11111111.11111111.11000000	172.19.1.128/26	62
3	General Lab 2	10101100.00010011.00000001.11000000 / 11111111.11111111.11111111.11000000	172.19.1.192/26	62

4	Cisco Lab	10101100.00010011.00000010.00000000 / 11111111.11111111.11111111.11000000	172.19.2.0/26	62
5	Embedded Lab	10101100.00010011.00000010.01000000 / 11111111.11111111.11111111.11000000	172.19.2.64/26	62
6	Network Devices	10101100.00010011.00000010.10000000 / 11111111.11111111.11111111.11100000	172.19.2.128/27	30
7	Video Conference Room	10101100.00010011.00000010.10100000 / 11111111.11111111.11111111.11110000	172.19.2.160/28	14

5.2 Range of IP address, Network and Broadcast address for the subnet

Area	Subnet	Network Address	Usable IP range	Broadcast address
Student Lounge(WiFi)	172.19.0.0/24	172.19.0.0	172.19.0.1-172.19.0.254	172.19.0.255
Hybrid Classroom	172.19.1.0/25	172.19.1.0	172.19.1.1-172.19.1.126	172.19.1.127
General Lab1	172.19.1.128/26	172.19.1.128	172.19.1.129-172.19.1.190	172.19.1.191
General Lab2	172.19.1.192/26	172.19.1.192	172.19.1.193-172.19.1.254	172.19.1.255
Cisco Lab	172.19.2.0/26	172.19.2.0	172.19.2.1-172.19.2.62	172.19.2.63
Embedded Lab	172.19.2.64/26	172.19.2.64	172.19.2.65-172.19.2.126	172.19.2.127
Network Devices	172.19.2.128/27	172.19.2.128	172.19.2.128-172.19.2.158	172.19.2.159
Video Conference Room	172.19.2.160/28	172.19.2.160	172.19.2.161-172.19.2.174	172.19.2.175

5.3 Range of IP address for user types based on the area

Student Lounge

Subnet: **172.19.0.0/24**

Device	**IP Address**
Wireless Access Points	172.19.0.1 - 172.19.0.10
Student Devices	172.19.0.11 – 172.19.0.254

Hybrid Classroom

Subnet: **172.19.1.0/25**

Device	**IP Address**
Lecturer PC	172.19.1.1
Smart Board	172.19.1.2
Projector	172.19.1.3
Camera & Audio Devices	172.19.1.4 - 172.19.1.10
Student Devices	172.19.1.11 - 172.19.1.126

General Lab 1

Subnet: **172.19.1.128/26**

Device	**IP Address**
Lecturer (PC)	172.19.1.129
Student (PCs)	172.19.1.130 – 172.19.1.160

General Lab 2

Subnet: **172.19.1.192/26**

Devices	**IP Address**
Lecturer (PC)	172.19.1.193
Student (PCs)	172.19.1.194 – 172.19.1.223

Cisco Lab

Subnet: **172.19.2.0/26**

Devices	**IP Address**
Lecturer (PC)	172.19.2.1
Student (PCs)	172.19.2.2 – 172.19.2.31
Routers & Switches	172.19.2.32 - 172.19.2.55

Embedded Lab

Subnet: **172.19.2.64/26**

Devices	**IP Address**
Lecturer (PC)	172.19.2.65
Student (PCs)	172.19.2.66 – 172.19.2.95
IoT / Embedded Devices	172.19.2.96 - 172.19.2.110

Video Conference Room

Subnet: **172.19.2.160/28**

Devices	**IP Address**
Conference PC	172.19.2.161
Camera	172.19.2.162
Projector	172.19.2.163
Display Screen	172.19.2.164

Network Devices

Subnet: **172.19.2.128/27**

Devices	**IP Address**
Core Switch	172.19.2.129
Access Switches	172.19.2.130 – 172.19.2.140
Wireless Access Points	172.19.2.141 - 172.19.2.150
Router	172.19.2.151 - 172.19.2.158

Reflection Task 5

Through Task 5, we gained an understanding of subnetting and IP address allocation in an academic network environment. This task required us to carefully analyze the given network address and divide it into appropriate subnets based on different areas such as laboratories, classrooms, and shared spaces. By completing this task, we reflected on the importance of accurate subnet mask selection, correct calculation of network and broadcast addresses, and efficient use of available IP addresses. The process helped us realize that proper planning is important to ensure that each area has sufficient IP addresses while avoiding unnecessary address wastage.

This task also allowed us to reflect on how theoretical networking concepts are applied in real-world scenarios. Assigning IP ranges based on user types and device functions helped us better understand how networks are structured to support different activities and requirements. We learned that even small calculation errors can affect the entire network design.

4.0 Conclusion

The project is fully completed within a few weeks of work and tasks. Throughout this project, we faced a lot of problems and doubts when it came to designing a network. Although it was quite tough for us, we still managed to overcome the problems and complete all the tasks including fixing some errors we made while checking before submitting. The project that we made achieved the goals that we set for each of the tasks, and from this project, we learned a lot of knowledge and gained experience in designing and building a network system that can be proposed in real life. In fact, we learned how to research and make decisions on the appropriate network devices based on the given Budget and requirements. This will be our first experience designing a LAN system, including the floor plan drawing and determining the best position for each device. Moreover, we also learned how to set up the IP address scheme for every network in a system. We believe that this experience will definitely come in handy for us to design a network system again in the future.

Besides the knowledge of implementing a network system, we also learned soft skills, which is a team working, which can only be gained when it comes to teamwork. We noticed the importance of teamwork when doing this project as we might have made some mistakes throughout this project and we will discuss and point out our doubts on certain points we do not understand. After this project, we believed that communication within the team was the main key to completing a group project.

In conclusion, we appreciate this opportunity to design a network system to gain experience and apply the knowledge learned from this course. We believe that the knowledge is beneficial in this project or even in the future, and without the bits of knowledge learned in this course, the project might not be done with the requirements mentioned.

5.0 Team Members and Responsibilities

Team Members	Responsibilities
Sarah Hamizah	Task 1: Design the ground floor and make changes on first floor Task 2: Prepare questions and provide answer with proper citation and answer operational feasibility. Task 3: Elaborate reasons why each device is chosen based on performance, reliability, future scalability and price. Supervises all modifications made by team members and finalises the report edits. Task 4:Identify cable types and cable lengths along with identifying work areas on the ground floor and first floor according to the approved floor plan. Task 5: Checking the outcome based on the topology and examining the IP address chosen. Task 6:
Nurul Huda Nabilah	
Luo Cimei	
Amr Khaled	

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25.

Appendices

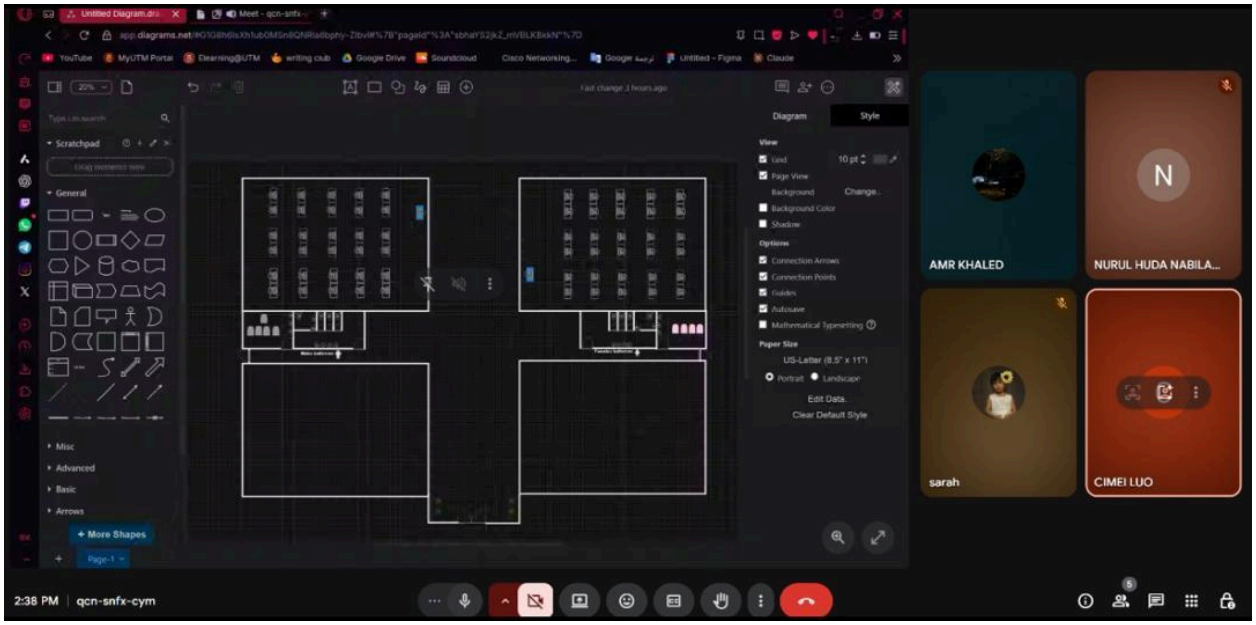
- The financial budget (what is bought, how much money is used and left, etc)

Details	Quantity	Amount (RM)
Income		
Sponsor from Universiti Teknologi Malaysia		1,600 000.00
Expenses		
Router (Cisco ISR 4321)	9	42876.99
Core Switches(Cisco Catalyst 9300)	2	30000
Access Switch (Cisco Catalyst 9200)	7	68740
Wireless Access Point(Cisco 1850 / 9115AX)	13	28691
Firewall(Cisco Firepower 1010 NGFW)	1	10000
Camera(HikvisionDS-2CD1043G2-LIU)	6	1974
Projector(Optoma HD28e)	6	17454
Screen(Manual pull-down screen)	6	1300.56
Printer (HP LaserJet Pro M404dn)	2	3000
IOT Devices(Raspberry Pi 4)	30	9000
Desktop computer(HP ProDesk 400 G9)	132	900240
Cable(		

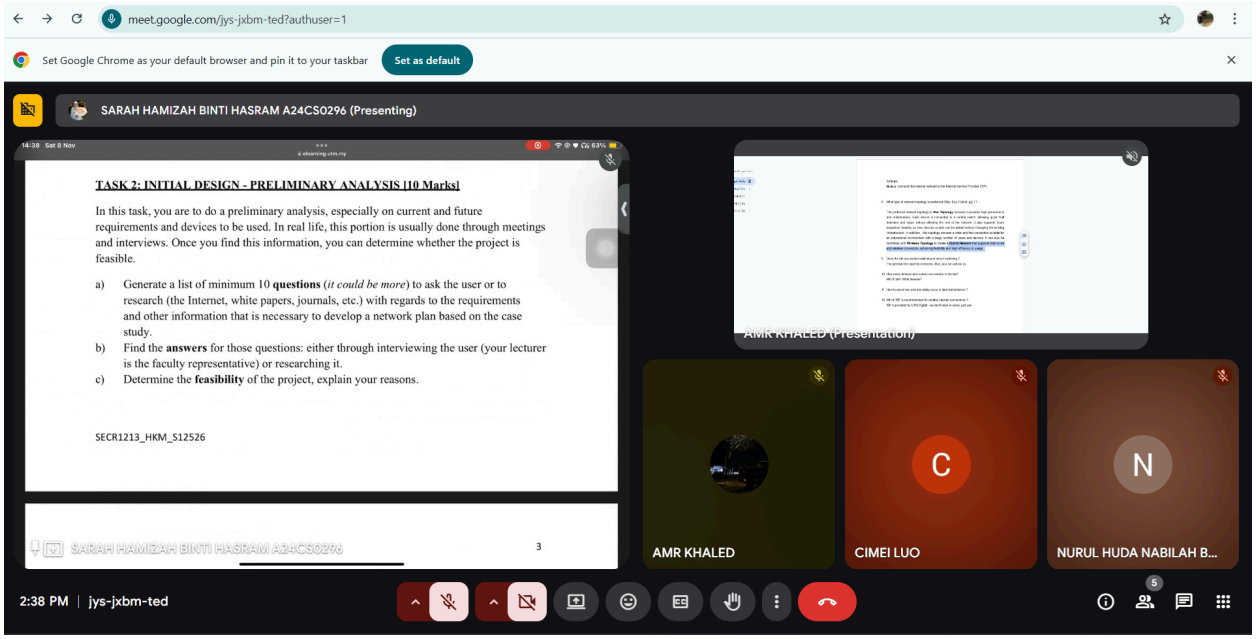
Total expenses		
Balance (income - total expenses)		

Rubric

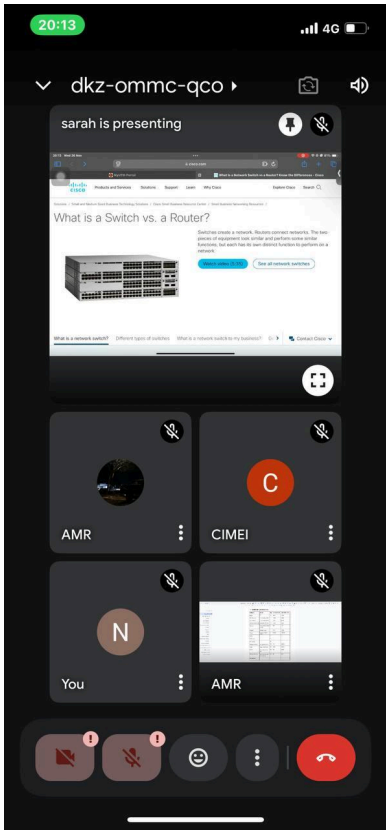
Pictures



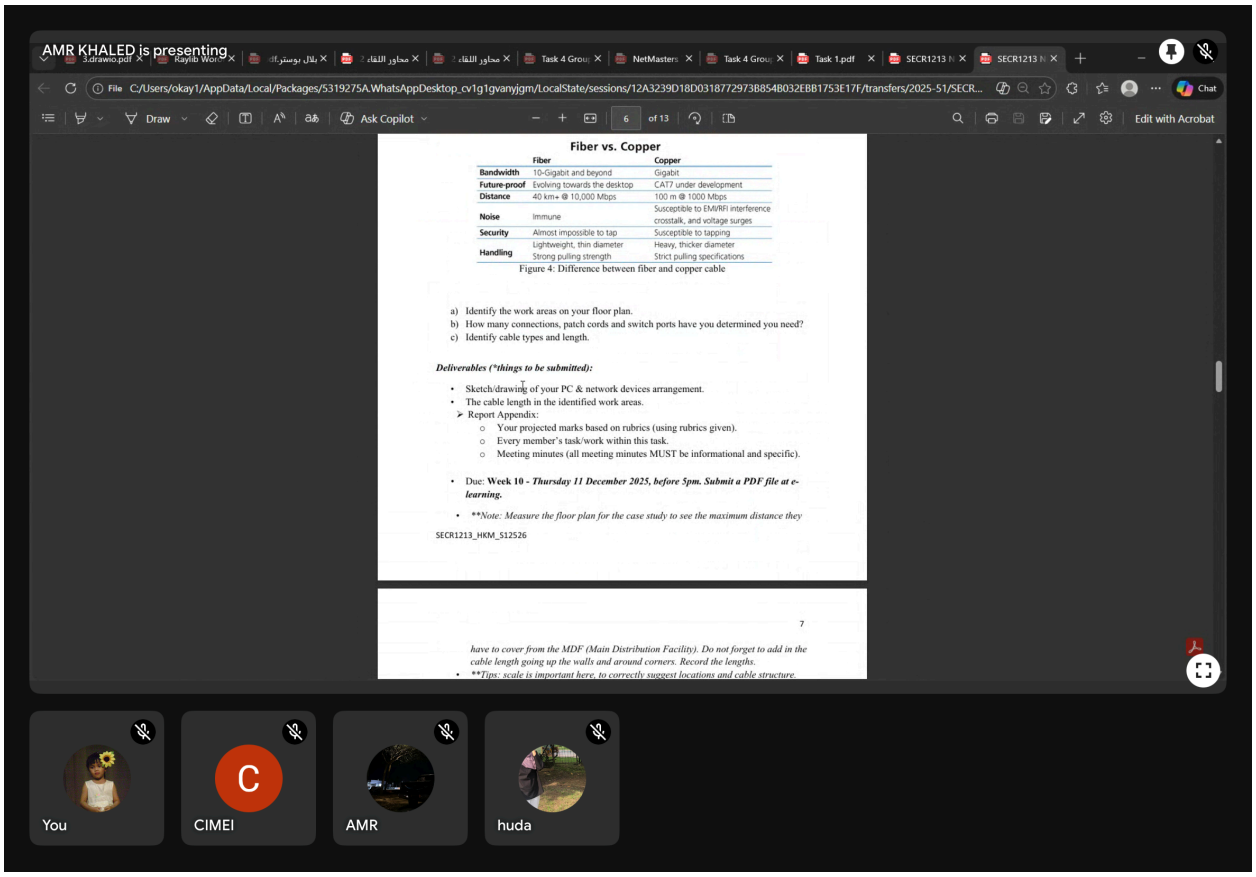
Task 1



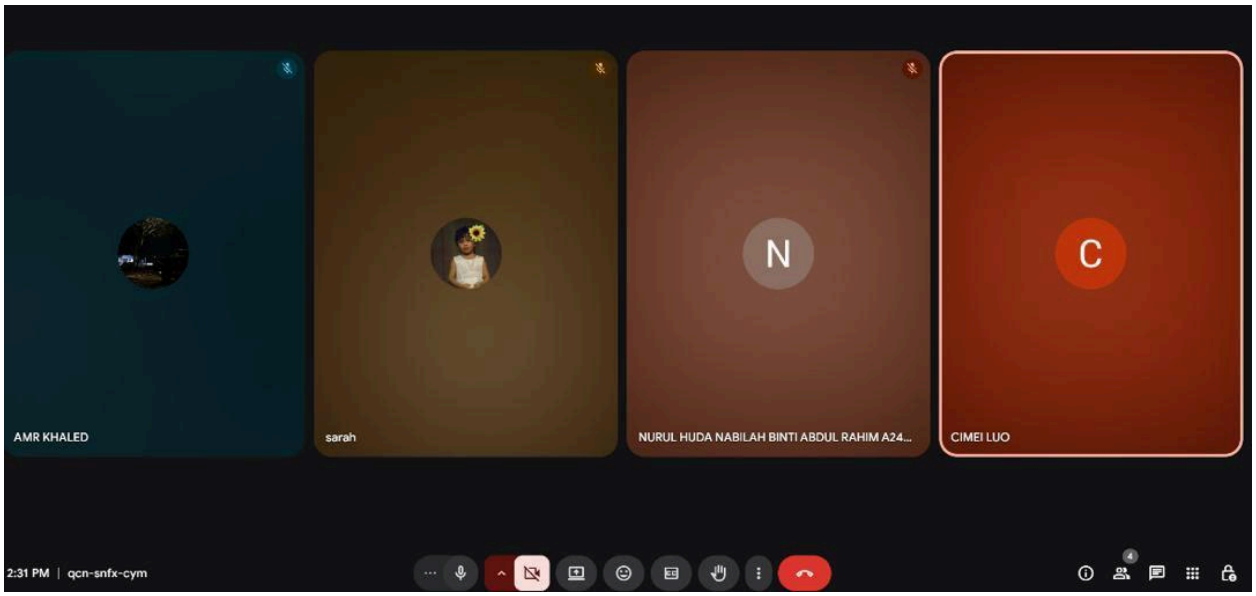
Task 2


20:13
4G
dkz-ommc-qco
sarah is presenting
What is a Switch vs. a Router?
AMR
C
CIMEI
N
You
AMR

Task 3

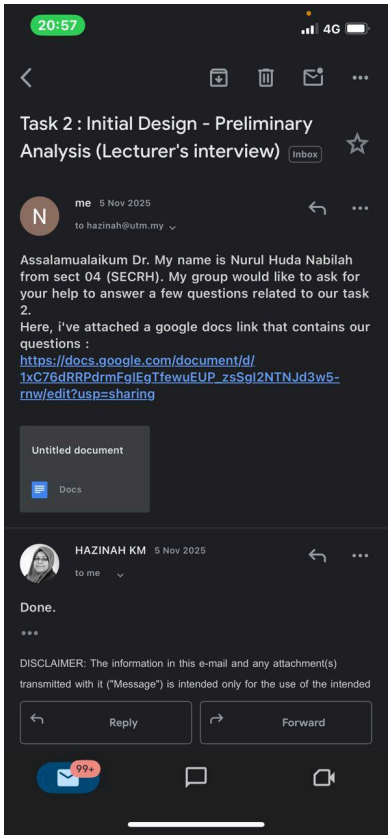

AMR KHALED is presenting
Draw
Ask Copilot
6
of 13
Edit with Acrobat
Fiber vs. Copper
Fiber
Copper
Bandwidth
10-Gigabit and beyond
Gigabit
Future-proof
Evolving towards the desktop
CAT7 under development
Distance
40 km+ @ 10,000 Mbps
100 m @ 1000 Mbps
Noise
Immune
Susceptible to EMI/RFI interference crosstalk, and voltage surges
Security
Almost impossible to tap
Susceptible to tapping
Handling
Lightweight, thin diameter
Strong pulling strength
Heavy, thicker diameter
Strict pulling specifications
Figure 4: Difference between fiber and copper cable
a) Identify the work areas on your floor plan.
b) How many connections, patch cords and switch ports have you determined you need?
c) Identify cable types and length.
Deliverables (*things to be submitted):
Sketch/drawing of your PC & network devices arrangement.
The cable length in the identified work areas.
Report Appendix:
Your projected marks based on rubrics (using rubrics given).
Every member's task/work within this task.
Meeting minutes (all meeting minutes MUST be informational and specific).
Due: Week 10 - Thursday 11 December 2025, before 5pm. Submit a PDF file at e-learning.
**Note: Measure the floor plan for the case study to see the maximum distance they
SECR1213_HKM_S12526
7
have to cover from the MDF (Main Distribution Facility). Do not forget to add in the cable length going up the walls and around corners. Record the lengths.
**Tips: scale is important here, to correctly suggest locations and cable structure.
You
CIMEI
AMR
huda

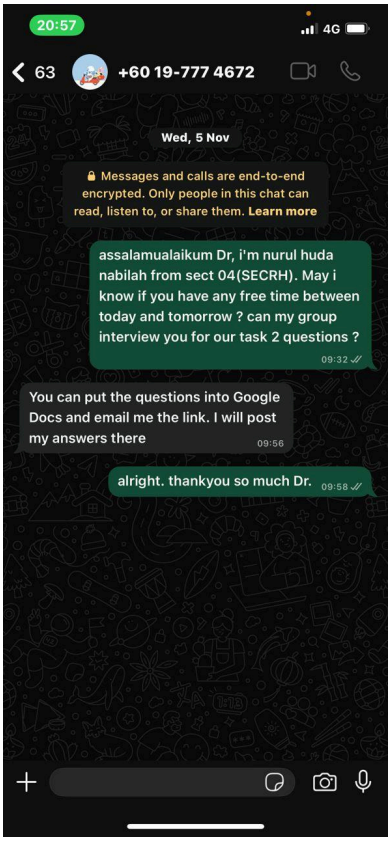
Task 4



figureTask 5 meeting

People we met to discuss the project





Meeting Minutes Task 1

Date / Time	14 October 2025 / 1400
Location	Google Meet
Agenda	Distribution of workloads and brainstorming ideas
Meeting MC	Luo Cimei

Attendance

Name	**Time**	**Reason for Absence**
Sarah Hamizah	1400	-
Nurul Huda Nabilah	1400	-
Luo Cimei	1400	-
Amr	1400	-

Minutes

No.	**Item Discussed**	**Ideas and the Person Giving It**	**Person in Charge & Date**
1	Find the website used for the sketch	Amr Khaled - Suggested using the draw.io website to design and sketch the floor plan. - General lab on the second floor.	Amr (14/10)
2	First floor design	Sarah Hamizah - Design the student lounge, video conference room, prayer room, and hybrid classroom.	Sarah (14/10)
3	Second floor design	Luo Cimei - Design the embedded lab and Cisco lab.	Cimei (14/10)

4	Final edits of the report	Huda Nabilah - Supervises all modifications made by team members and finalises the report edits	Huda (14/10)
5	Meeting ended	1600	

Meeting Minutes Task 2

Date / Time	3 November 2025 / 1400
Location	Google Meet
Agenda	1. Understand the details about task 2
	2. Suggest and select possible questions for task 2
Meeting MC	Nurul Huda Nabilah

Attendance

Name	**Time**	**Reason for Absence**
Sarah Hamizah	1400	-
Nurul Huda Nabilah	1400	-
Luo Cimei	1400	-
Amr	1400	-

Minutes

No.	**Item Discussed**	**Ideas and the Person Giving It**	**Person in Charge & Date**
1	Determine the feasibility	Luo Cimei - Proposed dividing feasibility into four main categories which is schedule, economic, operational and technical feasibility. - Each aspect was assign to one member.	Cimei (3/11)
2	Minimum number of question to ask	Amr Khaled - Suggested preparing at least 12 questions - Proposed that each member should be responsible for	Amr (3/11)

		preparing a minimum of 3 questions.	
3	Decide who needs to interview	Huda Nabilah - Will conduct an interview with lecturer (faculty representative) to clarify user requirement and technical needs - Other member focus on online research	Huda (3/11)
4	Report structure	Sarah Hamizah - Reminded the team that each member should prepare their assigned questions and answer - Responsible for compiling and editing the final report before submission.	Sarah (3/11)
5	Meeting ended	1600	

Meeting Minutes Task 3

Date / Time	24 November 2025 / 2000
Location	Google Meet
Agenda	1. Choose a suitable LAN devices
	2. Talk about the budget given
Meeting MC	Sarah Hamizah Binti Hasram

Attendance		
Name	**Time**	**Reason for Absence**
Sarah Hamizah	2000	-
Nurul Huda Nabilah	2000	-
Luo Cimei	2000	-
Amr	2000	-

Minutes

No.	**Item Discussed**	**Ideas and the Person Giving It**	**Person in Charge & Date**
1	Identify devices needed for the network	Amr Khaled - Suggested listing devices based on each lab and room - Suggested and comparing some devices to make sure a suitable device is selected	Amr (24/11)
2	Research device brands and model	Luo Cimei - Do a lot of research about the brand and model that are available in Malaysia - Recommended checking Malaysian pricing in Senheng, VST ECS and Ingram Micro	Luo Cimei (24/11)

3	Determine quantity of devices and check budget	Nurul Huda Nabilah - Suggested calculating devices based on PC x Labs - Recommended creating a spreadsheet to total the cost for all devices and compare the assigned budget.	Nurul Huda Nabilah (24/11)
4	Final edits of the report and answer the reflection question	Sarah Hamizah - Suggested writing justification based on performance, reliability, future scalability and price - Supervises all modifications made by team members and finalises the report edits.	Sarah Hamizah (24/11)
5	Meeting ended	2200	

Meeting Minutes Task 4

Date / Time	18 December 2025 / 1400
Location	Google Meet
Agenda	Distribution of task 4
	Analysis Task 4 requirement
Meeting MC	Amr Khaled

Attendance		
Name	**Time**	**Reason for Absence**
Sarah Hamizah	1400	-
Nurul Huda Nabilah	1400	-
Luo Cimei	1400	-
Amr	1400	-

Minutes			
No.	**Item Discussed**	**Ideas and the Person Giving It**	**Person in Charge & Date**
1	Understanding Task 4 requirement	Amr Khaled - Discussed the objectives of Task 4, focusing on identifying work areas, planning cabling layout, and drawing network distribution diagrams based on the floor plan from Task 1. - Agreed that the explanation must clearly justify cabling types, switch placement, and connectivity design.	Amr (18/12)
2	Identify work area arrangement	Sarah Hamizah - Identified work areas on the ground floor and first floor according to the	Sarah (18/12)

		approved floor plan.	
3	Design the network distribution diagram	Luo Cimei - Proposed using a hierarchical star topology with a Main Distribution Facility (MDF) on the ground floor and an Intermediate Distribution Facility (IDF) on the first floor. - Suggested drawing separate network distribution diagrams for each floor for better clarity.	Cimei (18/12)
4	Research suitable type of cabling	Huda Nabilah - Suggested using CAT6 UTP cables for horizontal cabling within each floor due to high speed and cost efficiency. - Proposed using fiber optic cable as backbone cabling between floors to support high bandwidth and future expansion.	Huda (18/12)
5	Meeting ended		

Meeting Minutes Task 5

Date / Time	26 December 2025 / 1400
Location	Google Meet
Agenda	Distribution of task 5
	Analysis Task 5 requirement
Meeting MC	Sarah Hamizah

Attendance		
Name	**Time**	**Reason for Absence**
Sarah Hamizah	1400	-
Nurul Huda Nabilah	1400	-
Luo Cimei	1400	-
Amr	1400	-

Minutes			
No.	**Item Discussed**	**Ideas and the Person Giving It**	**Person in Charge & Date**
1	Determining Subnet Mask	Amr Khaled - Choosing the different subnet masks - Estimating the total number of IP addresses	Amr (26/12)
2	Evaluating the Network topology chosen	Sarah Hamizah - Checking the outcome based on the topology - Examining the IP address chosen	Sarah (26/12)
3	Range of IP address, Network and Broadcast address for the subnet	Luo Cimei - Determining which subnets to settle on	Cimei (26/12)
4	Range of IP address for user types based	Huda Nabilah - Determining IP	Huda (26/12)

	on the area	addresses for devices in each room	
5	Meeting ended	1530	